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FEATURES OF THIS EDITION INCLUDE:

Business Improvement (BI) theme connects statistical analysis and business decision making. BI ideas are highlighted liberally in the margins to identify instances when important business conditions have been identified.

Emphasis on the t -test versus t -decision in confidence intervals and hypothesis testing is now governed by a known/unknown rather than by sample size, reflecting current thinking and best practices in statistical education.

Regression testing is approached using a new stepped method—making the material easier to learn.

Regression has been simplified and is presented in a modular fashion, enabling more flexibility in the material covered and the sequence in which it is covered.

Advanced version of MegaStat, the Excel add-in program designed to optimize Excel for statistical analysis, is included on the Student CD. The authors acknowledge the importance of spreadsheets in statistical software by continuing to integrate Excel and the lab throughout the text.

For students and instructors who want to explore statistical concepts from a graphical perspective, *Visual Statistics* is included on the Student CD.



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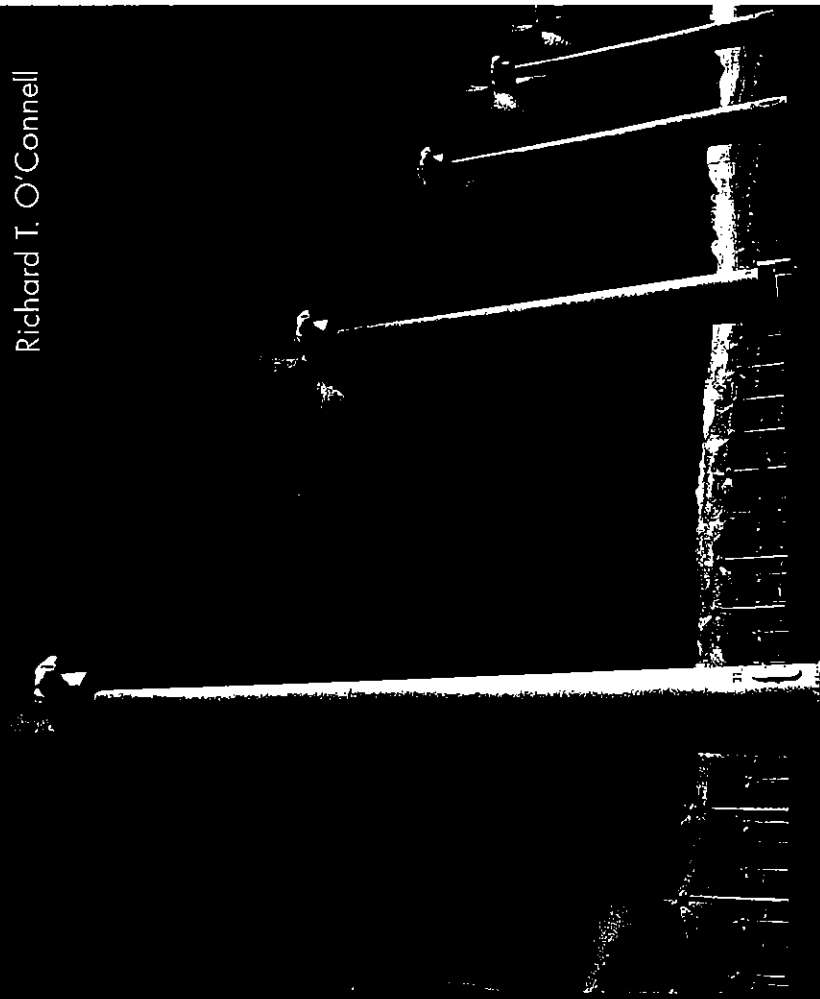
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Business Statistics in Practice

Fourth Edition

Bruce L. Bowerman
Richard T. O'Connell



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Discrete Random Variables

CHAPTER 4

Chapter Outline

- 4.1 Two Types of Random Variables
- 4.2 Discrete Probability Distributions

4.3 The Binomial Distribution

4.4 The Poisson Distribution (Optional)

W

When we observe an outcome of a random variable, we are observing a realization of the random variable. The outcome of a random variable is a numerical value. The outcome of a random variable is a numerical value. The outcome of a random variable is a numerical value.

4.1 Two Types of Random Variables

We begin with the definition of a random variable:

A random variable is a variable that assumes numerical values that are determined by the outcome of an experiment, where one and only one numerical value is assigned to each experimental outcome.

Before an experiment is carried out, its outcome is uncertain. It follows that, since a random variable assigns a number to each experimental outcome, a random variable can be thought of as representing an uncertain numerical outcome.

To illustrate the idea of a random variable, suppose that Sound City sells and installs car stereo systems. One of Sound City's most popular stereo systems is the TrueSound-XL, a top-of-the-line stereo cassette car radio. Consider the experiment of selling the TrueSound-XL radio at the Sound City store during a particular week. If we let x denote the number of radios sold during the week, then x is a random variable. That is, looked at before the week, the number of radios x that will be sold is uncertain, and, therefore, x is a random variable.

Notice that x , the number of TrueSound-XL radios sold in a week, might be 0 or 1 or 2 or 3, and so forth. In general, when the possible values of a random variable can be counted or listed, we say that the random variable is a discrete random variable. That is, either a discrete random variable may assume a finite number of possible values or the possible values may take the form of a countable sequence or list such as 0, 1, 2, 3, 4, ... (a countably infinite list).

Some other examples of discrete random variables are

- 1 The number, x , of the next three customers entering a store who will make a purchase. Here x could be 0, 1, 2, or 3.
- 2 The number, x , of four patients taking a new antibiotic who experience gastrointestinal distress as a side effect. Here x could be 0, 1, 2, 3, or 4.
- 3 The number, x , of television sets in a sample of 8 five-year-old television sets that have not needed a single repair. Here x could be any of the values 0, 1, 2, 3, 4, 5, 6, 7, or 8.
- 4 The rating, x , on a 1 through 5 scale given to a song by a listener in an AccuRatings music survey. Here x could be 1, 2, 3, 4, or 5.
- 5 The number, x , of major fires in a large city during the last two months. Here x could be 0, 1, 2, 3, and so forth (there is no definite maximum number of fires).
- 6 The number, x , of dirt specks in a one-square-yard sheet of plastic wrap. Here x could be 0, 1, 2, 3, and so forth (there is no definite maximum number of dirt specks).

The values of the random variables described in examples 1, 2, 3, and 4 are countable and finite. In contrast, the values of the random variables described in 5 and 6 are countable and infinite (or countably infinite lists). For example, in theory there is no limit to the number of major fires that could occur in a city in two months.

Not all random variables have values that are countable. When a random variable may assume any numerical value in one or more intervals on the real number line, then we say that the random variable is a continuous random variable.

Example 4.1 The Car Mileage Case

Consider the car mileage situation that we have discussed in Chapters 1 and 2. The EPA combined city and highway mileage, x , of a randomly selected midsize car is a continuous random variable. This is because, although we have measured mileages to the nearest one-tenth of a mile per gallon, technically speaking, the potential mileages that might be obtained correspond (starting

at, perhaps, 26 mpg) to an interval of numbers on the real line. We cannot count or list the numbers in such an interval because they are infinitesimally close together. That is, given any two numbers in an interval on the real line, there is always another number between them. To understand this, try listing the mileages starting with 26 mpg. Would the next mileage be 26.1 mpg? No, because we could obtain a mileage of 26.05 mpg. Would 26.05 mpg be the next mileage? No, because we could obtain a mileage of 26.025 mpg. We could continue this line of reasoning indefinitely. That is, whatever value we would try to list as the *next mileage*, there would always be another mileage between this *next mileage* and 26 mpg.

Some other examples of continuous random variables are

- 1 The temperature (in degrees Fahrenheit) of a cup of coffee served at a McDonald's restaurant.
- 2 The weight (in ounces) of strawberry preserves dispensed by an automatic filling machine into a 16-ounce jar.
- 3 The time (in minutes) that a customer in a store must wait to receive a credit card authorization.
- 4 The interest rate (in percent) charged for mortgage loans at a bank.

Exercises for Section 4.1

CONCEPTS

- 4.1 Explain the concept of a random variable.
- 4.2 Explain how the values of a discrete random variable differ from the values of a continuous random variable.
- 4.3 Classify each of the following random variables as discrete or continuous:
 - a x = the number of girls born to a couple who will have three children.
 - b x = the number of defects found on an automobile at final inspection.
 - c x = the weight (in ounces) of the sandwich meat placed on a submarine sandwich.
 - d x = the number of incorrect lab procedures conducted at a hospital during a particular week.
 - e x = the number of customers served during a given day at a drive-through window.
 - f x = the time needed by a clerk to complete a task.
 - g x = the temperature of a pizza oven at a particular time.

4.2 Discrete Probability Distributions

The value assumed by a discrete random variable depends on the outcome of an experiment. Because the outcome of the experiment will be uncertain, the value assumed by the random variable will also be uncertain. However, it is often useful to know the probabilities that are associated with the different values that the random variable can take on. That is, we often wish to know the random variable's probability distribution.

The probability distribution of a discrete random variable is a table, graph, or formula that gives the probability associated with each possible value that the random variable can assume.

We denote the probability distribution of the discrete random variable x as $p(x)$. As will be demonstrated in the following example, we can sometimes use the sample space of an experiment and probability rules to find the probability distribution of a random variable.

Example 4.2

Consider the pop quiz consisting of three true-false questions. Remember that the sample space when a student takes such a quiz consists of the outcomes

CCC CCI CCI ICC
CII CII CII III

We now define the random variable x to be the number of questions that the student answers correctly. Here x can assume the values 0, 1, 2, or 3. That is, the student could answer anywhere between 0 and 3 questions correctly. In Examples 3.3 and 3.5 we assumed that the

TABLE 4.1 Finding the Probability Distribution of x = the Number of Questions Answered Correctly When the Student Studies and Has a 90 Percent Chance of Answering Each Question Correctly

Value of x = the Number of Correct Answers	Sample Space Outcomes Corresponding to Value of x	Probability of Sample Space Outcome	$p(x)$ = Probability of the Value of x
$x = 0$ (no correct answers)	III	$(.1)(.1)(.1) = .001$	$p(0) = .001$
$x = 1$ (one correct answer)	CCI CIC ICI	$(.9)(.1)(.1) = .009$ $(.1)(.9)(.1) = .009$ $(.1)(.1)(.9) = .009$	$p(1) = .009 + .009 + .009 = .027$
$x = 2$ (two correct answers)	CCI CIC ICC	$(.9)(.9)(.1) = .081$ $(.9)(.1)(.9) = .081$ $(.1)(.9)(.9) = .081$	$p(2) = .081 + .081 + .081 = .243$
$x = 3$ (three correct answers)	CCC	$(.9)(.9)(.9) = .729$	$p(3) = .729$

student is totally unprepared for the quiz and thus has only a .5 probability of answering each question correctly. We now assume that the student studies and has a .9 probability of answering each question correctly. Table 4.1 summarizes finding the probabilities associated with each of the values of x (0, 1, 2, and 3). As an example of the calculations, consider finding the probability that x equals 2. Two questions will be answered correctly if and only if we obtain one of the sample space outcomes

CCI CCI ICC

Assuming that the three questions will be answered independently, these sample space outcomes have probabilities

$$P(CCI) = (.9)(.9)(.1) = .081$$

$$P(CIC) = (.9)(.1)(.9) = .081$$

$$P(ICC) = (.1)(.9)(.9) = .081$$

Therefore,

$$P(x = 2) = P(CCI) + P(CIC) + P(ICC)$$

$$= .081 + .081 + .081$$

$$= .243$$

Similarly, we can obtain probabilities associated with $x = 0$, $x = 1$, and $x = 3$. The probability distribution of x is summarized as follows:

x , Number of Questions Answered Correctly	$p(x)$, Probability of x
0	$p(0) = P(x = 0) = .001$
1	$p(1) = P(x = 1) = .027$
2	$p(2) = P(x = 2) = .243$
3	$p(3) = P(x = 3) = .729$

Notice that the probabilities in this probability distribution sum to $.001 + .027 + .243 + .729 = 1$.

To show the advantage of studying, note that the above probability distribution says that if the student has a .9 probability of answering each question correctly, then the probability that the student will answer all three questions correctly is .729. Furthermore, the probability that the student will answer at least two out of three questions correctly is (since the events $x = 2$ and $x = 3$ are mutually exclusive)

$$P(x \geq 2) = P(x = 2 \text{ or } x = 3)$$

$$= P(x = 2) + P(x = 3)$$

$$= .243 + .729$$

$$= .972$$

By contrast, we saw in Example 3.5 that if the student is totally unprepared and has only a .5 probability of answering each question correctly, then the probabilities that the student will

answer zero, one, two, and three questions correctly are, respectively, $1/8$, $3/8$, $3/8$, and $1/8$. Therefore, the probability that the unprepared student will answer all three questions correctly is only $1/8$, and the probability that this student will answer at least two out of three questions correctly is only $(3/8 + 1/8) = .5$.

In general, a discrete probability distribution $p(x)$ must satisfy two conditions:

Properties of a Discrete Probability Distribution $p(x)$

A discrete probability distribution $p(x)$ must be such that

$$1. \dots p(x) \geq 0 \text{ for each value of } x$$

$$2. \sum_{\text{all } x} p(x) = 1$$

The first of these conditions says that each probability in a probability distribution must be zero or positive. The second condition says that the probabilities in a probability distribution must sum to 1. Looking at the probability distribution illustrated in Example 4.2, we can see that these properties are satisfied.

Often it is not possible to examine the entire sample space of an experiment. In such a case we sometimes collect data that will allow us to estimate the probabilities in a probability distribution.

Example 4.3

Recall that Sound City sells the TrueSound-XL car radio, and define the random variable x to be the number of such radios sold in a particular week. In order to know the true probabilities of the various values of x , we would have to observe sales during all of the (potentially infinite number of) weeks in which the TrueSound-XL radio could be sold. That is, if we consider an experiment in which we randomly select a week and observe sales of the TrueSound-XL, the sample space would consist of a potentially infinite number of equally likely weeks. Obviously, it is not possible to examine this entire sample space.

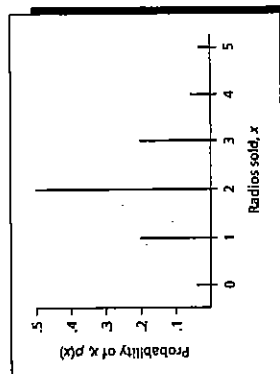
Suppose, however, that Sound City has kept historical records of TrueSound-XL sales during the last 100 weeks. These records tell us that

- 1 No radios have been sold in 3 (that is, $3/100 = .03$) of the weeks.
- 2 One radio has been sold in 20 (that is, $.20$) of the weeks.
- 3 Two radios have been sold in 50 (that is, $.50$) of the weeks.
- 4 Three radios have been sold in 20 (that is, $.20$) of the weeks.
- 5 Four radios have been sold in 5 (that is, $.05$) of the weeks.
- 6 Five radios have been sold in 2 (that is, $.02$) of the weeks.
- 7 No more than five radios were sold in any of the past 100 weeks.

It follows that we might estimate that the probability distribution of x , the number of TrueSound-XL radios sold during a particular week at Sound City, is as shown in Table 4.2. A graph of this distribution is shown in Figure 4.1.

x , Number of Radios Sold	$p(x)$, the Probability of x
0	$p(0) = P(X=0) = 3/100 = .03$
1	$p(1) = P(X=1) = 20/100 = .20$
2	$p(2) = P(X=2) = 50/100 = .50$
3	$p(3) = P(X=3) = 20/100 = .20$
4	$p(4) = P(X=4) = 5/100 = .05$
5	$p(5) = P(X=5) = 2/100 = .02$

FIGURE 4.1 A Graph of the Probability Distribution of x , the Number of TrueSound-XL Radios Sold at Sound City in a Week



Finally, it is reasonable to use the historical sales data from the past 100 weeks to estimate the true probabilities associated with the various numbers of radios sold if the sales process remains stable over time and is not seasonal (that is, if radio sales are not higher at one time of the year than at others).

Suppose that the experiment described by a random variable x is repeated on indefinitely large number of times. If the values of the random variable x observed on the repetitions are recorded, we would obtain the population of all possible observed values of the random variable x . This population has a mean, which we denote as μ_x , and which we sometimes call the expected value of x . In order to calculate μ_x , we multiply each value of x by its probability $p(x)$ and then sum the resulting products over all possible values of x .

The Mean or Expected Value of a Discrete Random Variable

The mean, or expected value, of a discrete random variable x is

$$\mu_x = \sum_{\text{all } x} xp(x)$$

In the next example we illustrate how to calculate μ_x , and we reason that the calculation really does give the mean of all possible observed values of the random variable x .

Example 4.4

Remember that Table 4.2 gives the probability distribution of x , the number of TrueSound-XL radios sold in a week at Sound City. Using this distribution, it follows that

$$\begin{aligned} \mu_x &= \sum_{\text{all } x} xp(x) \\ &= 0p(0) + 1p(1) + 2p(2) + 3p(3) + 4p(4) + 5p(5) \\ &= 0(.03) + 1(.20) + 2(.50) + 3(.20) + 4(.05) + 5(.02) \\ &= 2.1 \end{aligned}$$

To see that such a calculation gives the mean of all possible observed values of x , recall from Example 4.3 that the probability distribution in Table 4.2 was estimated from historical records of TrueSound-XL sales during the last 100 weeks. Also recall that these historical records tell us that during the last 100 weeks Sound City sold

- 1 Zero radios in 3 of the 100 weeks, for a total of $0(3) = 0$ radios
- 2 One radio in 20 of the 100 weeks, for a total of $1(20) = 20$ radios

- 3 Two radios in 50 of the 100 weeks, for a total of $2(50) = 100$ radios
 4 Three radios in 20 of the 100 weeks, for a total of $3(20) = 60$ radios
 5 Four radios in 5 of the 100 weeks, for a total of $4(5) = 20$ radios
 6 Five radios in 2 of the 100 weeks, for a total of $5(2) = 10$ radios

In other words, Sound City sold a total of

$$0 + 20 + 100 + 60 + 20 + 10 = 210 \text{ radios}$$

in 100 weeks, or an average of $210/100 = 2.1$ radios per week. Now, the average

$$\frac{210}{100} = \frac{0 + 20 + 100 + 60 + 20 + 10}{100}$$

can be written as

$$\frac{0(3) + 1(20) + 2(50) + 3(20) + 4(5) + 5(2)}{100}$$

which can be rewritten as

$$0\left(\frac{3}{100}\right) + 1\left(\frac{20}{100}\right) + 2\left(\frac{50}{100}\right) + 3\left(\frac{20}{100}\right) + 4\left(\frac{5}{100}\right) + 5\left(\frac{2}{100}\right) \\ = 0(.03) + 1(.20) + 2(.50) + 3(.20) + 4(.05) + 5(.02)$$

which equals $\mu_x = 2.1$. That is, if observed sales values occur with relative frequencies equal to those specified by the probability distribution in Table 4.2, then the average number of radios sold per week is equal to the expected value of x .

Of course, if we observe radio sales for another 100 weeks, the relative frequencies of the observed sales values would not (unless we are very lucky) be exactly as specified by the estimated probabilities in Table 4.2. Rather, the observed relative frequencies would differ somewhat from the estimated probabilities in Table 4.2, and the average number of radios sold per week would not exactly equal $\mu_x = 2.1$ (although the average would likely be close). However, the point is this: If the probability distribution in Table 4.2 were the true probability distribution of weekly radio sales, and if we were to observe radio sales for an indefinitely large number of weeks, then we would observe sales values with relative frequencies that are exactly equal to those specified by the probabilities in Table 4.2. In this case, when we calculate the expected value of x to be $\mu_x = 2.1$, we are saying that in the long run (that is, over an indefinitely large number of weeks) Sound City would average selling 2.1 TrueSound-XL radios per week.

As another example, again consider Example 4.2, and let the random variable x denote the number of the three true-false questions that the student who studies answers correctly. Using the probability distribution shown in Table 4.1, the expected value of x is

$$\mu_x = 0(.001) + 1(.027) + 2(.243) + 3(.729) \\ = 2.7$$

This expected value says that if a student takes a large number of three-question true-false quizzes and has a .9 probability of answering any single question correctly, then the student will average approximately 2.7 correct answers per quiz.

Example 4.5 The AccuRatings Case

In this example we will compute the popularity index for the song "Gangsta's Paradise" by Coolio. Recall from Example 3.16 (pages 148–151) that Strategic Radio Research had surveyed listeners rate this song as a 5, 4, 3, 2, 1, or UNFAM. Although not discussed in Example 3.16, Strategic Radio Research also estimated the proportions of listeners familiar with the song who would give the song ratings of 5, 4, 3, 2, and 1 to be, respectively, .43, .21, .22, .07, and .07. Now, it is reasonable to assign the numerical values 1 through 5 to the ratings 5 through 1 (this sort of thing is done when colleges assign the numerical values 4 through 0 to the grades A through F).

TABLE 4.3 An Estimate of the Probability Distribution of x , the Rating of the Song "Gangsta's Paradise" by a Randomly Selected Listener Who Is Familiar with This Song

x , Rating	$p(x)$, Probability of x
1	$p(1) = .07$
2	$p(2) = .07$
3	$p(3) = .22$
4	$p(4) = .21$
5	$p(5) = .43$

Therefore, we can regard the song's rating, x , by a randomly selected listener who is familiar with the song to be a discrete random variable having the estimated probability distribution shown in Table 4.3. It follows that the expected value of this estimated probability distribution is

$$\mu_x = 1(.07) + 2(.07) + 3(.22) + 4(.21) + 5(.43) \\ = 3.86$$

This estimated expected value is reported as the popularity index in Figure 3.8 (page 150) (the difference between the 3.86 calculated here and the 3.87 in Figure 3.8 is due to rounding). It says that Strategic Radio Research estimates that the mean rating of the song that would be given by all listeners who are familiar with the song is 3.86. As indicated in Figure 3.8, Strategic Radio Research reports that the song has a "very high popularity" index, which is the highest (#1) of all the songs rated for the week.

Example 4.6

An insurance company sells a \$20,000 whole life insurance policy for an annual premium of \$300. Actuarial tables show that a person who would be sold such a policy with this premium has a .001 probability of death during a year. Let x be a random variable representing the insurance company's profit made on one of these policies during a year. The probability distribution of x is

x , Profit	$p(x)$, Probability of x
\$300 (if the policyholder lives)	.999
\$300 - \$20,000 = -\$19,700 (a \$19,700 loss if the policyholder dies)	.001

The expected value of x (expected profit per year) is

$$\mu_x = \$300(.999) + (-\$19,700)(.001) \\ = \$280$$

This says that if the insurance company sells a very large number of these policies, it will average a profit of \$280 per policy per year. Since insurance companies actually do sell large numbers of policies, it is reasonable for these companies to make profitability decisions based on expected values.

Next, suppose that we wish to find the premium that the insurance company must charge for a \$20,000 policy if the company wishes the average profit per policy per year to be greater than \$0. If we let prem denote the premium the company will charge, then the probability distribution of the company's yearly profit x is

x , Profit	$p(x)$, Probability of x
prem (if policyholder lives)	.999
$\text{prem} - \$20,000$ (if policyholder dies)	.001

The expected value of x (expected profit per year) is

$$\mu_x = \text{prem}(.999) + (\text{prem} - 20,000)(.001) \\ = \text{prem} - 20$$



In order for this expected profit to be greater than zero, the premium must be greater than \$70. If, as previously stated, the company charges \$300 for such a policy, the \$280 charged in excess of the needed \$70 compensates the company for commissions paid to salespeople, administrative costs, dividends paid to investors, and other expenses.

In general, it is reasonable to base decisions on an expected value if we perform the experiment related to the decision (for example, if we sell the life insurance policy) many times. If we do not (for instance, if we perform the experiment only once), then it may not be a good idea to base decisions on the expected value. For example, it might not be wise for you—as an individual—to sell one person a \$20,000 life insurance policy for a premium of \$300. To see this, again consider the probability distribution of yearly profit:

x , Profit	$p(x)$, Probability of x
\$300 (if policyholder lives)	.999
\$300 - \$20,000 = -\$19,700 (if policyholder dies)	.001

and recall that the expected profit per year is \$280. However, since you are selling only one policy, you will not receive the \$280. You will either gain \$300 (with probability .999) or you will lose \$19,700 (with probability .001). Although the decision is personal, and although the chance of losing \$19,700 is very small, many people would not risk such a loss when the potential gain is only \$300.

Just as the population of all possible observed values of a discrete random variable x has a mean μ_x , this population also has a variance σ_x^2 and a standard deviation σ_x . Recall that the variance of a population is the average of the squared deviations of the different population values from the population mean. To find σ_x^2 , we calculate $(x - \mu_x)^2$ for each value of x , multiply $(x - \mu_x)^2$ by the probability $p(x)$, and sum the resulting products over all possible values of x .

The Variance and Standard Deviation of a Discrete Random Variable

The variance of a discrete random variable x is

$$\sigma_x^2 = \sum_{\text{all } x} (x - \mu_x)^2 p(x)$$

The standard deviation of x is the positive square root of the variance of x . That is,

$$\sigma_x = \sqrt{\sigma_x^2}$$

Example 4.7

Table 4.2 gives the probability distribution of x , the number of TrueSound-XL radios sold in a week at Sound City. Remembering that we have calculated μ_x (in Example 4.4) to be 2.1, it follows that

$$\begin{aligned}\sigma_x^2 &= \sum_{\text{all } x} (x - \mu_x)^2 p(x) \\ &= (0 - 2.1)^2 p(0) + (1 - 2.1)^2 p(1) + (2 - 2.1)^2 p(2) + (3 - 2.1)^2 p(3) \\ &\quad + (4 - 2.1)^2 p(4) + (5 - 2.1)^2 p(5) \\ &= (4.41)(.03) + (1.21)(.20) + (.01)(.50) + (3.61)(.05) + (8.41)(.02) \\ &= .89\end{aligned}$$

and that the standard deviation of x is $\sigma_x = \sqrt{.89} = .9434$.

The variance σ_x^2 and the standard deviation σ_x measure the spread of the population of all possible observed values of the random variable. To see how to use σ_x , remember that Chebyshev's

Theorem (see Chapter 2, page 74) tells us that, for any value of k that is greater than 1, at least $100(1 - 1/k^2)\%$ of all possible observed values of the random variable x lie in the interval $[\mu_x \pm k\sigma_x]$. Stated in terms of a probability, we have

$$P(x \text{ falls in the interval } [\mu_x \pm k\sigma_x]) \geq 1 - 1/k^2$$

For example, consider the probability distribution (in Table 4.2) of x , the number of TrueSound-XL radios sold in a week at Sound City. If we set k equal to 2, and if we use $\mu_x = 2.1$ and $\sigma_x = .9434$ to calculate the interval

$$\begin{aligned}[\mu_x \pm 2\sigma_x] &= [2.1 \pm 2(.9434)] \\ &= [.2132, 3.9868]\end{aligned}$$

then Chebyshev's Theorem tells us that

$$P(x \text{ falls in the interval } [.2132, 3.9868]) \geq 1 - 1/2^2 = 3/4$$

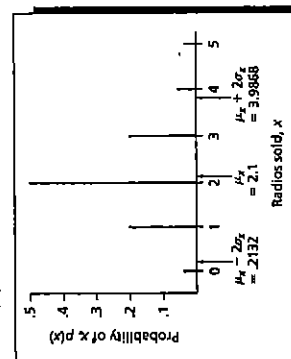
This says that in at least 75 percent of all weeks, Sound City will sell between .2132 and 3.9868 TrueSound-XL radios. As illustrated in Figure 4.2, there are three values of x between .2132 and 3.9868—namely, $x = 1$, $x = 2$, and $x = 3$. Therefore, the exact probability that x will be in the interval $[\mu_x \pm 2\sigma_x]$ is

$$p(1) + p(2) + p(3) = .20 + .50 + .20 = .90$$

This illustrates that, although Chebyshev's Theorem guarantees us that at least $100(1 - 1/k^2)\%$ of all possible observed values of a random variable x fall in the interval $[\mu_x \pm k\sigma_x]$, often the percentage is considerably higher.

In some cases, the graph of the probability distribution of a discrete random variable has the symmetrical, bell-shaped appearance of a normal curve. For example, the graph in Figure 4.2 is roughly bell-shaped and symmetrical. In such a situation—and under certain additional assumptions—the probability distribution can sometimes be approximated by a normal curve. We will discuss the needed assumptions in Chapter 5. As an example of such assumptions, note that although the graph in Figure 4.2 is roughly bell-shaped and symmetrical, it can be shown that there are not enough values of x , and thus not enough probabilities $p(x)$, for us to approximate the probability distribution by using a normal curve. If, however, the probability distribution of a discrete random variable x can be approximated by a normal curve, then the Empirical Rule for normally distributed populations describes the population of all possible values of x . Specifically, we can say that approximately 68.26 percent, 95.44 percent, and 99.73 percent of all possible observed values of x fall in the intervals $[\mu_x \pm \sigma_x]$, $[\mu_x \pm 2\sigma_x]$, and $[\mu_x \pm 3\sigma_x]$.

FIGURE 4.2 The Interval $[\mu_x \pm 2\sigma_x]$ for the Probability Distribution Describing TrueSound-XL Radio Sales (see Table 4.2)



To summarize, the standard deviation σ_x of a discrete random variable measures the spread of the population of all possible observed values of x . When the probability distribution of x can be approximated by a normal curve, this spread can be characterized by the Empirical Rule. When this is not possible, we can use Chebyshev's Theorem to characterize the spread of x .

To conclude this section, note that in Appendix D (Part 2) on page 850 we discuss various theoretical properties of the means and variances of random variables. In this appendix we also discuss the idea of the covariance between two random variables.

Exercises for Section 4.2

CONCEPTS

- 4.4 What is a discrete probability distribution? Explain in your own words.
- 4.5 What conditions must be satisfied by the probabilities in a discrete probability distribution? Explain what these conditions mean.
- 4.6 Describe how to compute the mean (or expected value) of a discrete random variable, and interpret what this quantity tells us about the observed values of the random variable.
- 4.7 Describe how to compute the standard deviation of a discrete random variable, and interpret what this quantity tells us about the observed values of the random variable.

METHODS AND APPLICATIONS

- 4.8 Explain whether each of the following is a valid probability distribution. If the probability distribution is valid, show why. Otherwise, show which condition(s) of a probability distribution are not satisfied.

x	$p(x)$	b	x	$p(x)$	c	x	$p(x)$	d	x	$p(x)$
-1	.2	1/2	-1	.1	2	.25	.1	2/7		
0	.6	3/4	0	.4	35	.7	4/7			
1	.2	1	2	.6	3	.9	1/7			

- 4.9 Consider each of the following probability distributions.

x	$p(x)$	b	x	$p(x)$	c	x	$p(x)$
0	.2	0	.25	-.2	.1		
1	.8	1	.45	0	.3		
		2	.2	2	.4		
		3	.1	5	.2		

Calculate μ_x and σ_x for each distribution. Then explain, using the probabilities, why μ_x is the mean of all possible observed values of x .

- 4.10 For each of the following, write out and graph the probability distribution of x . That is, list all the possible values of x and also list the corresponding probabilities. Then graph the distribution.

- a Refer to Exercise 3.3 (page 135), and let x equal the number of patrons who prefer diet cola A.
- b Refer to Exercise 3.4 (page 135), and let x equal the number of girls born to the couple.
- c Refer to Exercise 3.5 (page 135), and let x equal the number of people who will purchase a car.

- 4.11 For each of the following, find μ_x , σ_x^2 , and σ_x . Then interpret in words the meaning of μ_x and employ Chebyshev's rule to find intervals that contain at least $3/4$ and $8/9$ of the observed values of x .

- a x = the number of patrons who prefer diet cola A as defined in Exercise 4.10a.

- b x = the number of girls born to the couple as defined in Exercise 4.10b.

- c x = the number of people who will purchase a car as defined in Exercise 4.10c.

- 4.12 Suppose that the probability distribution of a random variable x can be described by the formula

$$p(x) = \frac{x}{15}$$

for each of the values $x = 1, 2, 3, 4$, and 5 . For example, then, $P(x = 2) = p(2) = 2/15$.

- a Write out the probability distribution of x .
- b Show that the probability distribution of x satisfies the properties of a discrete probability distribution.
- c Calculate the mean of x .
- d Calculate the variance, σ_x^2 , and the standard deviation, σ_x .

- 4.13 The following table summarizes investment outcomes and corresponding probabilities for a particular oil well:

x = the outcome in \$	$p(x)$
-\$40,000 (no oil)	.25
10,000 (some oil)	.7
70,000 (much oil)	.05

- a Graph $p(x)$; that is, graph the probability distribution of x .
- b Find the expected monetary outcome. Mark this value on your graph of part a. Then interpret this value.

- 4.14 In the book *Foundations of Financial Management* (7th ed.), Stanley B. Block and Geoffrey A. Hirt discuss risk measurement for investments. Block and Hirt present an investment with the possible outcomes and associated probabilities given in Table 4.4. The authors go on to say that the probabilities

may be based on past experience, industry ratios and trends, interviews with company executives, and sophisticated simulation techniques. The probability values may be easy to describe for the introduction of a mechanical sampling process in which the manufacturer has 10 years of past data, but difficult to assess for a new product in a foreign market.

- a Use the probability distribution in Table 4.4 to calculate the expected value (mean) and the standard deviation of the investment outcomes. Interpret the expected value.

- b Block and Hirt interpret the standard deviation of the investment outcomes as follows: "Generally, the larger the standard deviation (or spread of outcomes), the greater is the risk." Explain why this makes sense. Use Chebyshev's Theorem to illustrate your point.

- c Block and Hirt compare three investments having the following means and standard deviations of the investment outcomes:

Investment 1	Investment 2	Investment 3
$\mu = \$600$	$\mu = \$190$	$\mu = \$600$
$\sigma = \$20$	$\sigma = \$190$	$\sigma = \$300$

- Which of these investments involves the most risk? The least risk? Explain why by using Chebyshev's Theorem to compute an interval for each investment that will contain at least $8/9$ of the investment outcomes.

- d Block and Hirt continue by comparing two more investments:

Investment A	Investment B
$\mu = \$6,000$	$\mu = \$600$
$\sigma = \$600$	$\sigma = \$190$

The authors explain that Investment A

appears to have a high standard deviation, but not when related to the expected value of the distribution. A standard deviation of \$600 on an investment with an expected value of \$6,000 may indicate less risk than a standard deviation of \$190 on an investment with an expected value of only \$600.

We can eliminate the size difficulty by developing a third measure, the coefficient of variation (V). This term calls for nothing more difficult than dividing the standard deviation of an investment by the expected value. Generally, the larger the coefficient of variation, the greater is the risk.

$$\text{Coefficient of variation } (V) = \frac{\sigma}{\mu}$$

TABLE 4.4 Probability Distribution of Outcomes for an Investment

Outcome	Probability of Outcome	Assumptions
\$300	.2	Pessimistic
600	.6	Moderately successful
900	.2	Optimistic

Sources: S. B. Block and G. A. Hirt, *Foundations of Financial Management*, 7th ed., p. 378. Copyright © 1994. Reprinted by permission of McGraw-Hill Companies, Inc.

Calculate the coefficient of variation for investments A and B. Which investment carries the greater risk?

- e Calculate the coefficient of variation for investments 1, 2, and 3 in part c. Based on the coefficient of variation, which investment involves the most risk? The least risk? Do we obtain the same results as we did by comparing standard deviations (in part c)? Why?

4.15 An insurance company will insure a \$50,000 diamond for its full value against theft at a premium of \$400 per year. Suppose that the probability that the diamond will be stolen is .005, and let x denote the insurance company's profit.

- a Set up the probability distribution of the random variable x .
b Calculate the insurance company's expected profit.
c Find the premium that the insurance company should charge if it wants its expected profit to be \$1,000.

4.16 In the book *Foundations of Financial Management* (7th ed.), Stanley B. Block and Geoffrey A. Hirt discuss a semiconductor firm that is considering two choices: (1) expanding the production of semiconductors for sale to end users or (2) entering the highly competitive home computer market. The cost of both projects is \$60 million, but the net present value of the cash flows from sales and the risks are different.

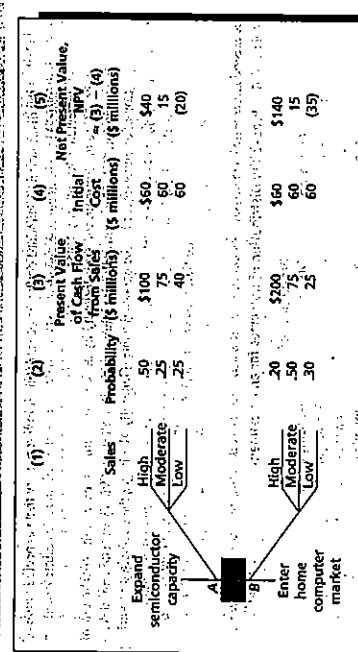
Figure 4.3 gives a tree diagram of the project choices. The tree diagram gives a probability distribution of expected sales for each project. It also gives the present value of cash flows from sales and the net present value (NPV = present value of cash flow from sales minus initial cost) corresponding to each sales alternative. Note that figures in parentheses denote losses.

- a For each project choice, calculate the expected net present value.
b For each project choice, calculate the variance and standard deviation of the net present value.
c Calculate the coefficient of variation for each project choice. See Exercise 4.14d for a discussion of the coefficient of variation.
d Which project has the higher expected net present value?
e Which project carries the least risk? Explain.
f In your opinion, which project should be undertaken? Justify your answer.

4.17 Five thousand raffle tickets are to be sold at \$10 each to benefit a local community group. The prizes, the number of each prize to be given away, and the dollar value of winnings for each prize are as follows:

Prize	Number to Be Given Away	Dollar Value
Automobile	1	\$13,000
Entertainment center	2	3,000 each
VCR	5	400 each
Gift certificate	50	20 each

FIGURE 4.3 A Tree Diagram of Two Project Choices



Source: S. B. Block and G. A. Hirt, *Foundations of Financial Management*, 7th ed., p. 367. Copyright © 1994. Reprinted by permission of McGraw-Hill Companies, Inc.

TABLE 4.5 Return Distributions for Companies A, B, and C and for Two Possible Acquisitions

Economic Condition	Probability	Company A Returns	Company B Returns	Company C Returns	Company A + B Returns	Company A + C Returns
1	2	17%	19%	13%	18%	15%
2	2	16	17	11	13	13
3	2	15	16	10	12	12
4	2	14	15	9	11	11
5	2	13	14	8	10	10

If you buy one ticket, calculate your expected winnings. (Form the probability distribution of x = your dollar winnings, and remember to subtract the cost of your ticket.)

4.18 Company A is considering the acquisition of two separate but large companies, Company B and Company C, having sales and assets equal to its own. Table 4.5 gives the probabilities of returns for each of the three companies under various economic conditions. The table also gives the probabilities of returns for each possible combination: Company A plus Company B, and Company A plus Company C.

- a For each of Companies A, B, and C find the mean return and the standard deviation of returns.
b Find the mean return and the standard deviation of returns for the combination of Company A plus Company B.
c Find the mean return and the standard deviation of returns for the combination of Company A plus Company C.
d Compare the mean returns for each of the two possible combinations—Company A plus Company B and Company A plus Company C. Is either mean higher? How do they compare to Company A's mean return?
e Compare the standard deviations of the returns for each of the two possible combinations—Company A plus Company B and Company A plus Company C. Which standard deviation is smaller? Which possible combination involves less risk? How does the risk carried by this combination compare to the risk carried by Company A alone?
f Which acquisition would you recommend—Company A plus Company B or Company A plus Company C?

4.19 THE ACCURATING CASE

Again consider Exercise 3.31 (page 153) and the title-by-title analysis of the song "We've Got It Goin' On" by the Backstreet Boys. Although not discussed in Exercise 3.31, Strategic Radio Research estimated the proportions of listeners familiar with the song who would give the song ratings of 5, 4, 3, 2, and 1 to be, respectively, .1978, .2967, .3077, .3879, and .1099. Assign the numerical values 1 through 5 to the ratings 1 through 5.

- a Find an estimate of the probability distribution of this song's rating, x , by a randomly selected listener who is familiar with the song.
b Find the popularity index for the song "We've Got It Goin' On" that would be reported by Strategic Radio Research. That is, find an estimate of the mean rating of this song that would be given by all listeners who are familiar with this song.

4.3 The Binomial Distribution

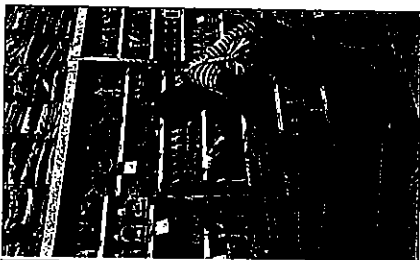
In this section we discuss what is perhaps the most important discrete probability distribution—the binomial distribution. We begin with an example.

Example 4.8

Suppose that historical sales records indicate that 40 percent of all customers who enter a discount department store make a purchase. What is the probability that two of the next three customers will make a purchase?

In order to find this probability, we first note that the experiment of observing three customers making a purchase decision has several distinguishing characteristics:

- The experiment consists of three identical trials; each trial consists of a customer making a purchase decision.



2 Two outcomes are possible on each trial: the customer makes a purchase (which we call a *success* and denote as S), or the customer does not make a purchase (which we call a *failure* and denote as F).

3 Since 40 percent of all customers make a purchase, it is reasonable to assume that $P(S)$, the probability that a customer makes a purchase, is .4 and is constant for all customers. This implies that $P(F)$, the probability that a customer does not make a purchase, is .6 and is constant for all customers.

4 We assume that customers make independent purchase decisions. That is, we assume that the outcomes of the three trials are independent of each other.

It follows that the sample space of the experiment consists of the following eight sample space outcomes:

SSS	FFS
SSF	FSF
SFS	SFF
FSS	FFF

Here the sample space outcome SSS represents all three customers making purchases. On the other hand, the sample space outcome FFF represents the first customer making a purchase, the second customer not making a purchase, and the third customer making a purchase.

Two out of three customers make a purchase if one of the sample space outcomes SSF, SFS, or FSS occurs. Furthermore, since the trials (purchase decisions) are independent, we can simply multiply the probabilities associated with the different trial outcomes (each of which is S or F) to find the probability of a sequence of outcomes:

$$\begin{aligned} P(SSF) &= P(S)P(S)P(F) = (.4)(.4)(.6) = (.4)^2(.6) \\ P(SFS) &= P(S)P(F)P(S) = (.4)(.6)(.4) = (.4)^2(.6) \\ P(FSS) &= P(F)P(S)P(S) = (.6)(.4)(.4) = (.4)^2(.6) \end{aligned}$$

It follows that the probability that two out of the next three customers make a purchase is

$$\begin{aligned} P(SSF) + P(SFS) + P(FSS) \\ &= (.4)^2(.6) + (.4)^2(.6) + (.4)^2(.6) \\ &= 3(.4)^2(.6) = .288 \end{aligned}$$

We can now generalize the previous result and find the probability that x of the next n customers will make a purchase. Here we will assume that p is the probability that a customer makes a purchase, $q = 1 - p$ is the probability that a customer does not make a purchase, and purchase decisions (trials) are independent. To generalize the probability that two out of the next three customers make a purchase, which equals

$$3(.4)^2(.6)$$

we note that

- 1 The 3 in this expression is the number of sample space outcomes (SSF, SFS, and FSS) that correspond to the event "two out of the next three customers make a purchase." Note that this number equals the number of ways we can arrange two successes among the three trials.
- 2 The .4 is p , the probability that a customer makes a purchase.
- 3 The .6 is $q = 1 - p$, the probability that a customer does not make a purchase.

Therefore, the probability that two of the next three customers make a purchase is

$$\left(\begin{array}{c} \text{The number of ways} \\ \text{to arrange 2 successes} \\ \text{among 3 trials} \end{array} \right) p^2 q^1$$

Now, notice that, although each of the sample space outcomes SSF, SFS, and FSS represents a different arrangement of the two successes among the three trials, each of these sample space outcomes consists of two successes and one failure. For this reason, the probability of each of these sample space outcomes equals $(.4)^2(.6)^1 = p^2 q^1$. It follows that p is raised to a power that equals the number of successes (2) in the three trials, and q is raised to a power that equals the number of failures (1) in the three trials.

In general, each sample space outcome describing the occurrence of x successes (purchases) in n trials represents a different arrangement of x successes in n trials. However, each outcome consists of x successes and $n - x$ failures. Therefore, the probability of each sample space outcome is $p^x q^{n-x}$. It follows by analogy that the probability that x of the next n trials are successes (purchases) is

$$\left(\begin{array}{c} \text{The number of ways} \\ \text{to arrange } x \text{ successes} \\ \text{among } n \text{ trials} \end{array} \right) p^x q^{n-x}$$

We can use the expression we have just arrived at to compute the probability of x successes in the next n trials if we can find a way to calculate the number of ways to arrange x successes among n trials. It can be shown that:

The number of ways to arrange x successes among n trials equals

$$\frac{n!}{x!(n-x)!}$$

where $n!$ is pronounced "n factorial" and is calculated as $n! = n(n-1)(n-2) \cdots (1)$ and where (by definition) $0! = 1$.

For instance, using this formula, we can see that the number of ways to arrange $x = 2$ successes among $n = 3$ trials equals

$$\frac{n!}{x!(n-x)!} = \frac{3!}{2!(3-2)!} = \frac{3!}{2!1!} = \frac{3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 1} = 3$$

Of course, we have previously seen that the three ways to arrange $x = 2$ successes among $n = 3$ trials are SSF, SFS, and FSS.

Using the preceding formula, we obtain the following general result:

The Binomial Distribution

A binomial experiment has the following characteristics:

- 1 The experiment consists of n identical trials.
- 2 Each trial results in a success or a failure.
- 3 The probability of a success on any trial is p and remains constant from trial to trial. This implies that the probability of failure, q , on any trial is $1 - p$ and remains constant from trial to trial.
- 4 The trials are independent (that is, the results of the trials have nothing to do with each other).

Furthermore, if we define the random variable

x = the total number of successes in n trials of a binomial experiment

then we call x a binomial random variable, and the probability of obtaining x successes in n trials is

$$p(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

To better understand these interpretations, we can use MINITAB to simulate the antibiotic to a large number of samples of four randomly selected patients. That is, we can use MINITAB to randomly select a large number of observations from the binomial distribution of Table 4.6. The MINITAB output of a simulation of 10,000 samples of four randomly selected patients is given in Figure 4.4(c). In other words, this figure gives the results obtained when MINITAB has randomly selected 10,000 observations from the binomial distribution with $p = .10$ and $n = 4$. Each observation (or simulated sample of four patients) results in either 0, 1, 2, 3, or 4 patients experiencing nausea. The MINITAB output presents the results in the form of a histogram that shows the number of simulated samples in which 0, 1, 2, 3, or 4 patients experienced nausea. For instance, none of the four patients experienced nausea in 6,544 (that is, in 65.44 percent) of the 10,000 samples, while 3 out of 4 patients experienced nausea in 26 (or in .26 percent) of the 10,000 samples. These simulated results are quite close to the percentages given by the probabilities in Table 4.6. If we could use MINITAB to simulate an indefinitely large number of samples of four patients, then the simulated percentages would be exactly equal to the percentages given by the binomial probabilities. Note, of course, that we cannot do this—we can simulate only some large number of samples (say, 10,000 samples).

Another way to avoid hand calculating binomial probabilities is to use binomial tables, which have been constructed to give the probability of x successes in n trials. A table of binomial probabilities is given in Table A.1 (page 817). A portion of this table is reproduced in Table 4.7(a) and (b). Part (a) of this table gives binomial probabilities corresponding to $n = 4$ trials. Values of p , the probability of success, are listed across the top of the table (ranging from $p = .05$ to $p = .50$ in steps of .05), and more values of p (ranging from $p = .30$ to $p = .95$ in steps of .05) are listed across the bottom of the table. When the value of p being considered is one of those across the top of the table, for instance, to find the probabilities (in four trials) are listed down the left side of the table. For values of x (the number of successes in four trials) are listed down the top of the table. For instance, to find the probabilities that we have computed in Table 4.6, we look in part (a) of Table 4.7 ($n = 4$) and read down the column labeled .10. Remembering that the values of x are on the left side of the table because $p = .10$ is on top of the table, we find the probabilities in Table 4.6 (they are shaded). For example, the probability that none of four

TABLE 4.7
Binomial Probabilities

		Values of p (.05 to .50)									
		.05	.10	.15	.20	.25	.30	.35	.40	.45	.50
Number of Successes	0	.8145	.6810	.5623	.4596	.3664	.2816	.2041	.1365	.0769	.0439
	1	.1715	.3170	.3770	.3770	.3170	.2816	.2041	.1365	.0769	.0439
	2	.0375	.1488	.2376	.2936	.3170	.2816	.2041	.1365	.0769	.0439
	3	.0025	.0130	.0270	.0439	.0769	.1365	.2041	.2816	.3170	.3770
	4	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
		Values of p (.50 to .95)									
		.50	.60	.70	.80	.90	.95	.975	.990	.995	.999
Number of Successes	0	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	1	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	2	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	3	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	4	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000

(b) A Table for $n = 8$ trials

		Values of p (.05 to .50)									
		.05	.10	.15	.20	.25	.30	.35	.40	.45	.50
Number of Successes	0	.6830	.4305	.2725	.1678	.1001	.0576	.0319	.0168	.0084	.0039
	1	.2733	.3826	.3947	.3355	.2670	.1977	.1373	.0896	.0548	.0313
	2	.0515	.1488	.2376	.2936	.3170	.2816	.2041	.1365	.0769	.0439
	3	.0025	.0130	.0270	.0439	.0769	.1365	.2041	.2816	.3170	.3770
	4	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
		Values of p (.50 to .95)									
		.50	.60	.70	.80	.90	.95	.975	.990	.995	.999
Number of Successes	0	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	1	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	2	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	3	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
	4	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000

patients experience nausea is $p(0) = .6561$, the probability that one of the four patients experiences nausea is $p(1) = .2916$, and so forth. If the value of p is across the bottom of the table, then we read the values of x from the right side of the table. As an example, if p equals .60, then the probability of two successes in four trials is $p(2) = .3456$ (we have shaded this probability).

Example 4.11

Suppose that we wish to investigate whether p , the probability that a patient will experience nausea as a side effect of taking Phe-Mycin, is greater than .10, the maximum value of p claimed by the drug company. This assessment will be made by assuming, for the sake of argument, that p equals .10, and by using sample information to weigh the evidence against this assumption and in favor of the conclusion that p is greater than .10. Suppose that when a sample of $n = 4$ randomly selected patients is treated with Phe-Myacin, three of the four patients experience nausea. Since the fraction of patients in the sample that experience nausea is $3/4 = .75$, which is far greater than .10, we have some evidence contradicting the assumption that p equals .10. To evaluate the strength of this evidence, we calculate the probability that at least 3 out of 4 randomly selected patients would experience nausea as a side effect if, in fact, p equals .10. Using the binomial probabilities in Table 4.7(a), and realizing that the events $x = 3$ and $x = 4$ are mutually exclusive, we have

$$\begin{aligned} P(x \geq 3) &= P(x = 3 \text{ or } x = 4) \\ &= P(x = 3) + P(x = 4) \\ &= .0036 + .0001 \\ &= .0037 \end{aligned}$$

This probability says that, if p equals .10, then in only .37 percent of all possible samples of four randomly selected patients would at least three of the four patients experience nausea as a side effect. This implies that, if we are to believe that p equals .10, then we must believe that we have observed a sample result that is so rare that it can be described as a 37 in 10,000 chance. Because observing such a result is very unlikely, we have very strong evidence that p does not equal .10 and is, in fact, greater than .10.

Next, suppose that we consider what our conclusion would have been if only one of the four randomly selected patients had experienced nausea. Since the sample fraction of patients who experienced nausea is $1/4 = .25$, which is greater than .10, we would have some evidence to contradict the assumption that p equals .10. To evaluate the strength of this evidence, we calculate the probability that at least one out of four randomly selected patients would experience nausea as a side effect of being treated with Phe-Myacin if, in fact, p equals .10. Using the binomial probabilities in Table 4.7(a), we have

$$\begin{aligned} P(x \geq 1) &= P(x = 1 \text{ or } x = 2 \text{ or } x = 3 \text{ or } x = 4) \\ &= P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4) \\ &= .2916 + .0486 + .0036 + .0001 \\ &= .3439 \end{aligned}$$

This probability says that, if p equals .10, then in 34.39 percent of all possible samples of four randomly selected patients, at least one of the four patients would experience nausea. Since it is not particularly difficult to believe that a 34.39 percent chance has occurred, we would not have much evidence against the claim that p equals .10.

Example 4.11 illustrates what is sometimes called the *rare event approach* to making a statistical inference. The idea of this approach is that if the probability of an observed sample result under a given assumption is small, then we have strong evidence that the assumption is false. Although there are no strict rules, many statisticians judge the probability of an observed sample result to be small if it is less than .05. The logic behind this will be explained more fully in Chapter 8.

Example 4.12

The manufacturer of the ColorSmart-5000 television set claims that 95 percent of its sets last at least five years without requiring a single repair. Suppose that we contact $n = 8$ randomly selected ColorSmart-5000 purchasers five years after they purchased their sets. Each purchaser's set will have needed no repairs (a success) or will have been repaired at least once (a failure). We will assume that p , the true probability that a purchaser's television set will require no repairs within five years, is .95, as claimed by the manufacturer. Furthermore, it is reasonable to believe that the repair records of the purchasers' sets are independent of each other. Let x denote the number of the $n = 8$ randomly selected sets that have lasted at least five years without a single repair. Then x is a binomial random variable that can take on any of the potential values 0, 1, 2, 3, 4, 5, 6, 7, or 8. The binomial distribution of x is listed in Table 4.8. Here we have obtained these probabilities from Table 4.7(b). To use the table, we look at the column corresponding to $p = .95$. Because $p = .95$ is listed at the bottom of the table, we read the values of x and their corresponding probabilities from bottom to top (we have shaded the probabilities). Notice that the values of x are listed on the right side of the table.

Figure 4.5(a) gives the MINITAB output of the binomial distribution with $p = .95$ and $n = 8$ (that is, the binomial distribution of Table 4.8). This binomial distribution is graphed in

$$P(x) = \frac{n!}{x!(n-x)!} (.95)^x (.05)^{n-x}$$

x , Number of Sets That Require No Repairs	$P(x)$
0	$P(0) = .0000$
1	$P(1) = .0000$
2	$P(2) = .0000$
3	$P(3) = .0000$
4	$P(4) = .0004$
5	$P(5) = .0054$
6	$P(6) = .0515$
7	$P(7) = .2793$
8	$P(8) = .6634$

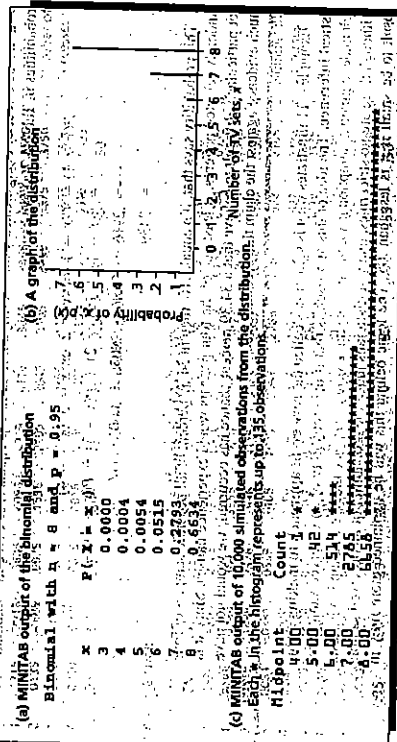


Figure 4.5(b), and Figure 4.5(c) gives the MINITAB output of 10,000 simulated observations from this distribution. Looking at Figure 4.5(c), we see that, for example, seven out of eight sets have lasted at least five years without a single repair in 2,785 (27.85 percent) of the 10,000 simulated samples. This result is very close to the percentage (27.93 percent) given by the binomial distribution [see Figure 4.5(a)].

Next, suppose that when we actually contact eight randomly selected purchasers, we find that five out of the eight television sets owned by these purchasers have lasted at least five years without a single repair. Since the sample fraction, $5/8 = .625$, of television sets needing no repairs is less than .95, we have some evidence contradicting the manufacturer's claim that p equals .95. To evaluate the strength of this evidence, we will calculate the probability that five or fewer of the eight randomly selected televisions would last five years without a single repair if, in fact, p equals .95. Using the binomial probabilities in Table 4.8, we have

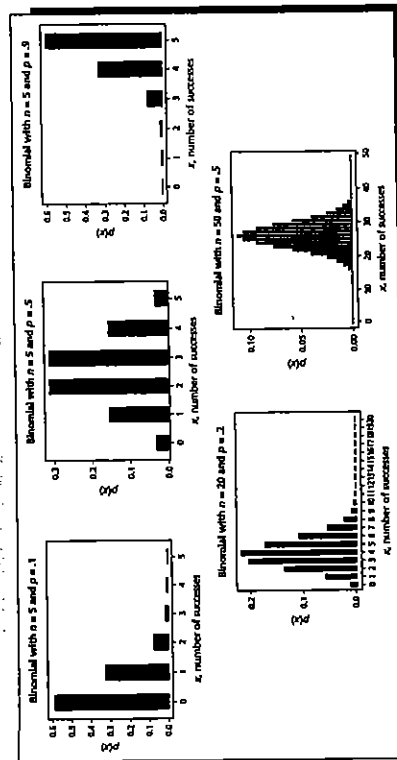
$$\begin{aligned}
 P(x \leq 5) &= P(x = 5 \text{ or } x = 4 \text{ or } x = 3 \text{ or } x = 2 \text{ or } x = 1 \text{ or } x = 0) \\
 &= P(x = 5) + P(x = 4) + P(x = 3) + P(x = 2) + P(x = 1) + P(x = 0) \\
 &= .0054 + .0004 + .0000 + .0000 + .0000 + .0000 \\
 &= .0058
 \end{aligned}$$

This probability says that, if p equals .95, then in only .58 percent of all possible samples of eight randomly selected ColorSmart-5000 televisions would five or fewer of the eight televisions last five years without a single repair. Therefore, if we are to believe that p equals .95, we must believe that a 58 in 10,000 chance has occurred. Since it is difficult to believe that such a small chance has occurred, we have strong evidence that p does not equal .95, and is, in fact, less than .95.

In Examples 4.10 and 4.12 we have illustrated binomial distributions with different values of n and p . The values of n and p are often called the parameters of the binomial distribution. Figure 4.6 shows several different binomial distributions. We see that, depending on the parameters, a binomial distribution can be skewed to the right, skewed to the left, or symmetrical.

We next consider calculating the mean, variance, and standard deviation of a binomial random variable. If we place the binomial probability formula into the expressions (given in Section 4.2) for the mean and variance of a discrete random variable, we can derive formulas that allow us to

FIGURE 4.6 Several Binomial Distributions



easily compute μ_x , σ_x^2 , and σ_x for a binomial random variable. Omitting the details of the derivation, we have the following results:

The Mean, Variance, and Standard Deviation of a Binomial Random Variable
If x is a binomial random variable, then

$$\mu_x = np, \quad \sigma_x^2 = npq, \quad \sigma_x = \sqrt{npq}$$

where n is the number of trials, p is the probability of success on each trial, and $q = 1 - p$ is the probability of failure on each trial.

As a simple example, again consider the television manufacturer, and recall that x is the number of eight randomly selected ColorSmart-5000 televisions that last five years without a single repair. If the manufacturer's claim that p equals .95 is true (which implies that q equals $1 - p = 1 - .95 = .05$), it follows that

$$\begin{aligned}\mu_x &= np = 8(.95) = 7.6 \\ \sigma_x^2 &= npq = 8(.95)(.05) = .38 \\ \sigma_x &= \sqrt{npq} = \sqrt{.38} = .6164\end{aligned}$$

In order to interpret $\mu_x = 7.6$, suppose that we were to randomly select all possible samples of eight ColorSmart-5000 televisions and record the number of sets in each sample that last five years without a repair. If we averaged all of our results, we would find that the average number of sets per sample that last five years without a repair is equal to 7.6.

To conclude this section, note that in Appendix C (Part 2) on page 846 we discuss the hypergeometric distribution. This distribution is related to the binomial distribution. The main difference between the two distributions is that in the case of the hypergeometric distribution, the trials are not independent and the probabilities of success and failure change from trial to trial. This occurs when we sample without replacement from a finite population. However, when the finite population is large compared to the sample, the binomial distribution can be used to approximate the hypergeometric distribution. The details are explained in Appendix C (Part 2).

Exercises for Section 4.3

CONCEPTS

- 4.20 List the four characteristics of a binomial experiment.
4.21 Suppose that x is a binomial random variable. Explain what the values of x represent. That is, how are the values of x defined?
4.22 Explain the logic behind the rare event approach to making statistical inferences.

METHODS AND APPLICATIONS

- 4.23 Suppose that x is a binomial random variable with $n = 5$, $p = .3$, and $q = .7$.
a Write the binomial formula for this situation and list the possible values of x .
b For each value of x , calculate $P(x)$, and graph the binomial distribution.
c Find $P(x = 3)$.
d Find $P(x \leq 3)$.
e Find $P(x < 3)$.
f Find $P(x \geq 4)$.
g Find $P(x > 2)$.
h Use the probabilities you computed in part b to calculate the mean, μ_x , the variance, σ_x^2 , and the standard deviation, σ_x , of this binomial distribution. Show that the formulas for μ_x , σ_x^2 , and σ_x given in this section give the same results.
i Calculate the interval $[\mu_x \pm 2\sigma_x]$. Use the probabilities of part b to find the probability that x will be in this interval.
4.24 Thirty percent of all customers who enter a store will make a purchase. Suppose that six customers enter the store and that these customers make independent purchase decisions.
a Let x = the number of the six customers who will make a purchase. Write the binomial formula for this situation.

- b Use the binomial formula to calculate
(1) The probability that exactly five customers make a purchase.
(2) The probability that at least three customers make a purchase.
(3) The probability that two or fewer customers make a purchase.
(4) The probability that at least one customer makes a purchase.
4.25 The customer service department for a wholesale electronics outlet claims that 90 percent of all customer complaints are resolved to the satisfaction of the customer. In order to test this claim, a random sample of 13 customers who have filed complaints is selected.
a Let x = the number of sampled customers whose complaints were resolved to the customer's satisfaction. Assuming the claim is true, write the binomial formula for this situation.
b Use the binomial tables (see Table A.1, page 817) to find each of the following if we assume that the claim is true:
(1) $P(x \leq 13)$.
(2) $P(x > 10)$.
(3) $P(x \geq 14)$.
(4) $P(9 \leq x \leq 12)$.
(5) $P(x \leq 9)$.

- c Suppose that 2 of the 13 customers selected, 9 have had their complaints resolved satisfactorily. Using part b, do you believe the claim of 90 percent satisfaction? Explain.

4.26 The United States Golf Association requires that the weight of a golf ball must not exceed 1.62 oz. The association periodically checks golf balls sold in the United States by sampling specific brands stocked by pro shops. Suppose that a manufacturer claims that no more than 1 percent of its brand of golf balls exceed 1.62 oz. in weight. Suppose that 24 of this manufacturer's golf balls are randomly selected, and let x denote the number of the 24 randomly selected golf balls that exceed 1.62 oz. Figure 4.7 gives part of a MegaStat output of the binomial distribution with $n = 24$, $p = .01$, and $q = .99$. (Note that, since $P(X = x) = .0000$ for values of x from 6 to 24, we omit these probabilities.) Use this output to

- a Find $P(x = 0)$, that is, find the probability that none of the randomly selected golf balls exceeds 1.62 oz. in weight.
b Find the probability that at least one of the randomly selected golf balls exceeds 1.62 oz. in weight.
c Find $P(x \leq 3)$.
d Find $P(x \geq 2)$.
e Suppose that 2 of the 24 randomly selected golf balls are found to exceed 1.62 oz. Using your result from part d, do you believe the claim that no more than 1 percent of this brand of golf balls exceed 1.62 oz. in weight?

4.27 An industry representative claims that 50 percent of all satellite dish owners subscribe to at least one premium movie channel. In an attempt to justify this claim, the representative will poll a randomly selected sample of dish owners.

- a Suppose that the representative's claim is true, and suppose that a sample of four dish owners is randomly selected. Assuming independence, use an appropriate formula to compute
(1) The probability that none of the dish owners in the sample subscribes to at least one premium movie channel.
(2) The probability that more than two dish owners in the sample subscribe to at least one premium movie channel.
b Suppose that the representative's claim is true, and suppose that a sample of 20 dish owners is randomly selected. Assuming independence, what is the probability that
(1) Nine or fewer dish owners in the sample subscribe to at least one premium movie channel?
(2) More than 11 dish owners in the sample subscribe to at least one premium movie channel?
(3) Fewer than five dish owners in the sample subscribe to at least one premium movie channel?
c Suppose that, when we survey 20 randomly selected dish owners, we find that 4 of the dish owners actually subscribe to at least one premium movie channel. Using a probability you found in this exercise as the basis for your answer, do you believe the industry representative's claim? Explain.

4.28 For each of the following, calculate μ_x , σ_x^2 , and σ_x by using the formulas given in this section. Then interpret the meaning of μ_x , σ_x^2 , and σ_x .
(1) Interpret the probability that x falls in the interval $[\mu_x \pm 2\sigma_x]$.
a The situation of Exercise 4.24, where x = the number of the six customers who will make a purchase.

FIGURE 4.7
MegaStat Output
of the Binomial
Distribution with
 $n = 24$, $p = .01$
and $q = .99$

x	$P(X)$
0	0.78538
1	0.19047
2	0.02213
3	0.00164
4	0.00009
5	0.00000

- b The situation of Exercise 4.25, where x = the number of 15 sampled customers whose complaints were resolved to the customer's satisfaction.
- c The situation of Exercise 4.26, where x = the number of the 24 randomly selected golf balls that exceed 1.62 oz. in weight.
- 4.29 The January 1986 mission of the Space Shuttle Challenger was the 25th such shuttle mission. It was unsuccessful due to an explosion caused by an O-ring seal failure.
- a According to NASA, the probability of such a failure in a single mission was 1/60,000. Using this value of p and assuming all missions are independent, calculate the probability of no mission failures in 25 attempts. Then calculate the probability of at least one mission failure in 25 attempts.
- b According to a study conducted for the Air Force, the probability of such a failure in a single mission was 1/35. Recalculate the probability of no mission failures in 25 attempts and the probability of at least one mission failure in 25 attempts.
- c Based on your answers to parts a and b, which value of p seems more likely to be true? Explain.
- d How small must p be made in order to ensure that the probability of no mission failures in 25 attempts is .999?

4.4 The Poisson Distribution (Optional) ●●●

We now discuss a discrete random variable that describes the number of occurrences of an event over a specified interval of time or space. For instance, we might wish to describe (1) the number of customers who arrive at the checkout counters of a grocery store in one hour, or (2) the number of major fires in a city during the last two months, or (3) the number of dirt specks found in one square yard of plastic wrap.

Such a random variable can often be described by a Poisson distribution. We describe this distribution and give two assumptions needed for its use in the following box:

The Poisson Distribution

Consider the number of times an event occurs over an interval of time or space, and assume that

1. The probability of the event's occurrence is the same for any two intervals of equal length, and
2. Whether the event occurs in any interval is independent of whether the event occurs in any other nonoverlapping interval.

Then, the probability that the event will occur x times in a specified interval is

$$p(x) = \frac{e^{-\mu} \mu^x}{x!}$$

Here μ is the mean (or expected) number of occurrences of the event in the specified interval, and $e = 2.71828 \dots$ is the base of Napierian logarithms.

In theory, there is no limit to how large x might be. That is, theoretically speaking, the event under consideration could occur an indefinitely large number of times during any specified interval. This says that a Poisson random variable might take on any of the values 0, 1, 2, 3, . . . and so forth. We will now look at an example.

Example 4.13

In an article² in the August 15, 1998, edition of *The Journal News* (Hamilton, Ohio), the Associated Press reported that the Cleveland Air Route Traffic Control Center, the busiest in the nation for guiding planes on cross-country routes, had experienced an unusually high number of errors since the end of July. An error occurs when controllers direct flights either within five miles of each other horizontally, or within 2,000 feet vertically at a height of 18,000 feet or more (the standard is 1,000 feet vertically at heights less than 18,000 feet). The controllers' union blamed

² F. J. Frommer, "Errors on the Rise at Traffic Control Center in Ohio," *The Journal News*, August 15, 1998.

TABLE 4.9 A Portion of a Poisson Probability Table

x, Number of Occurrences	μ , Mean Number of Occurrences									
	.1	.2	.3	.4	.5	.6	.7	.8	.9	1.0
0	.9048	.8187	.7408	.6703	.6065	.5488	.4966	.4493	.4066	.3679
1	.0955	.1637	.2222	.2831	.3393	.3930	.4438	.4917	.5367	.5793
2	.0045	.0164	.0333	.0536	.0758	.0988	.1217	.1458	.1697	.1939
3	.0002	.0011	.0026	.0047	.0072	.0100	.0128	.0157	.0184	.0211
4	.0000	.0001	.0003	.0007	.0010	.0016	.0020	.0027	.0031	.0035
5	.0000	.0000	.0000	.0001	.0002	.0004	.0007	.0010	.0012	.0015
6	.0000	.0000	.0000	.0000	.0000	.0000	.0001	.0002	.0003	.0005

x, Number of Occurrences	μ , Mean Number of Occurrences									
	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	.3329	.3012	.2725	.2466	.2231	.2019	.1827	.1653	.1496	.1353
1	.3662	.3614	.3543	.3452	.3347	.3230	.3106	.2975	.2842	.2707
2	.2014	.2169	.2303	.2417	.2510	.2584	.2640	.2678	.2700	.2707
3	.0738	.0867	.0998	.1128	.1255	.1378	.1496	.1607	.1710	.1804
4	.0203	.0260	.0324	.0395	.0471	.0551	.0636	.0723	.0812	.0902
5	.0045	.0062	.0084	.0111	.0141	.0176	.0216	.0260	.0309	.0361
6	.0008	.0012	.0018	.0026	.0035	.0047	.0061	.0078	.0098	.0120
7	.0001	.0002	.0003	.0005	.0008	.0011	.0015	.0020	.0027	.0034
8	.0000	.0000	.0001	.0001	.0001	.0002	.0003	.0005	.0006	.0009

Source: From Brookdale © 1991.

the errors on a staff shortage, whereas the Federal Aviation Administration (FAA) claimed that the cause was improved error reporting and an unusual number of thunderstorms.

Suppose that an air traffic control center has been averaging 20.8 errors per year and that the center experiences 3 errors in a week. The FAA must decide whether this occurrence is unusual enough to warrant an investigation as to the causes of the (possible) increase in errors. To investigate this possibility, we will find the probability distribution of x , the number of errors in a week, when we assume that the center is still averaging 20.8 errors per year.

Arbitrarily choosing a time unit of one week, the average (or expected) number of errors per week is $20.8/52 = 4$. Therefore, we can use the Poisson formula (note that the Poisson assumptions are probably satisfied) to calculate the probability of no errors in a week to be

$$p(0) = P(x = 0) = \frac{e^{-\mu} \mu^0}{0!} = \frac{e^{-4} (4)^0}{1} = .6703$$

Similarly, the probability of three errors in a week is

$$p(3) = P(x = 3) = \frac{e^{-\mu} \mu^3}{3!} = \frac{e^{-4} (4)^3}{3 \cdot 2 \cdot 1} = .0072$$

As with the binomial distribution, tables have been constructed that give Poisson probabilities. A table of these probabilities is given in Table A.2 (page 821). A portion of this table is reproduced in Table 4.9. In this table, values of the mean number of occurrences, μ , are listed across the top of the table, and values of x (the number of occurrences) are listed down the left side of the table. In order to use the table in the traffic control situation, we look at the column in Table 4.9 corresponding to 4, and we find the probabilities of 0, 1, 2, 3, 4, 5, and 6 errors (we have shaded these probabilities). For instance, the probability of one error in a week is .2681. Also, note that the probability of any number of errors greater than 6 is so small that it is not listed in the table. Table 4.10 summarizes the Poisson distribution of x , the number of errors in a week. This table also shows how the probabilities associated with the different values of x are calculated.

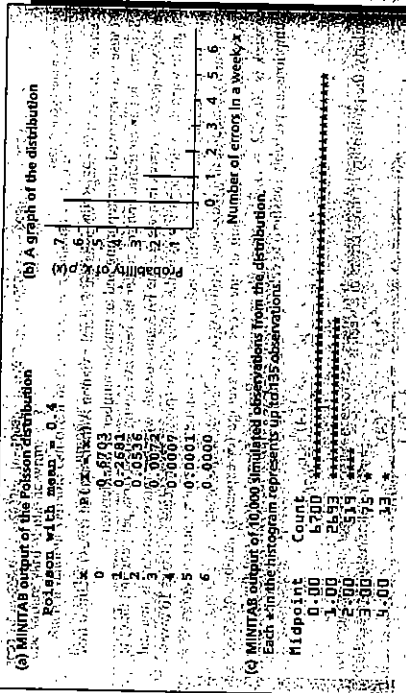
Poisson probabilities can also be calculated by using MINITAB, Excel, and MegaStat. For instance, Figure 4.8(a) gives the MINITAB output of the Poisson distribution presented in Table 4.10.³ This Poisson distribution is graphed in Figure 4.8(b), and Figure 4.8(c) gives the

³As we will show in the appendices to this chapter, we can use Excel and MegaStat to obtain output of the Poisson distribution that is essentially identical to the output given by MINITAB.

4.9 The Poisson Distribution of Errors in a Week

x , the Number of Errors in a Week

x	$p(x) = \frac{e^{-\mu} \mu^x}{x!}$
0	$p(0) = \frac{e^{-4} 4^0}{0!} = .0703$
1	$p(1) = \frac{e^{-4} 4^1}{1!} = .2681$
2	$p(2) = \frac{e^{-4} 4^2}{2!} = .0536$
3	$p(3) = \frac{e^{-4} 4^3}{3!} = .0072$
4	$p(4) = \frac{e^{-4} 4^4}{4!} = .0007$
5	$p(5) = \frac{e^{-4} 4^5}{5!} = .0001$
6	$p(6) = \frac{e^{-4} 4^6}{6!} = .0000$



MINITAB output of a histogram of 10,000 simulated observations from this distribution. Looking at Figure 4.8(c), we see that, for example, three errors have occurred in 75 (.75 percent) of the 10,000 simulated weeks. This result is very close to the Percentage (.72 percent) given by the Poisson distribution [see Figure 4.8(b)].

Next, recall that there have been three errors at the air traffic control center in the last week. This is considerably more errors than 4, the expected number of errors assuming the center is still averaging 20.8 errors per year. Therefore, we have some evidence to contradict this assumption. To evaluate the strength of this evidence, we calculate the probability that at least three errors will occur in a week if, in fact, μ equals 4. Using the Poisson probabilities in Table 4.10 (for $\mu = 4$), we obtain

$$P(x \geq 3) = p(3) + p(4) + p(5) + p(6) = .0072 + .0007 + .0001 + .0000 = .008$$

This probability says that, if the center is averaging 20.8 errors per year, then there would be three errors in a week in only .8 percent of all weeks. That is, if we are to believe that the control center

is averaging 20.8 errors per year, then we must believe that an 8 in 1,000 chance has occurred. Since it is very difficult to believe that such a rare event has occurred, we have strong evidence that the average number of errors per week has increased. Therefore, an investigation by the FAA into the reasons for such an increase is probably justified.

Example 4.14

In the book *Modern Statistical Quality Control and Improvement*, Nicholas R. Farnum (1994) presents an example dealing with the quality of computer software. In the example, Farnum measures software quality by monitoring the number of errors per 1,000 lines of computer code.

Suppose that the number of errors per 1,000 lines of computer code is described by a Poisson distribution with a mean of four errors per 1,000 lines of code. If we wish to find the probability of obtaining eight errors in 2,500 lines of computer code, we must adjust the mean of the Poisson distribution. To do this, we arbitrarily choose a *space unit* of one line of code, and we note that a mean of four errors per 1,000 lines of code is equivalent to $4/1,000$ of an error per line of code. Therefore, the mean number of errors per 2,500 lines of code is $(4/1,000)(2,500) = 10$. It follows that

$$p(8) = \frac{e^{-\mu} \mu^8}{8!} = \frac{e^{-10} 10^8}{8!} = .1126$$

The mean, μ , is often called the *parameter* of the Poisson distribution. Figure 4.9 shows several Poisson distributions. We see that, depending on its parameter (mean), a Poisson distribution can be very skewed to the right or can be quite symmetrical.

Finally, if we place the Poisson probability formulas into the general expressions (of Section 4.2) for μ , σ^2 , and σ , we can derive formulas for calculating the mean, variance, and standard deviation of a Poisson distribution:

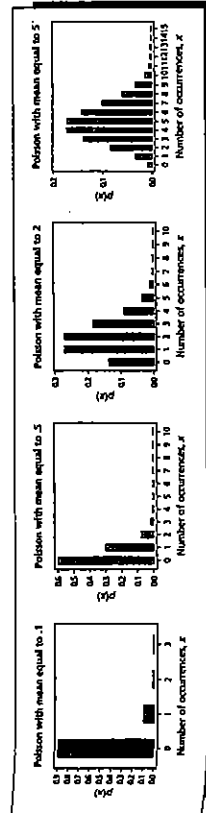
The Mean, Variance, and Standard Deviation of a Poisson Random Variable

Suppose that x is a Poisson random variable. If μ is the average number of occurrences of an event over the specified interval of time or space of interest, then

$$\mu_x = \mu \quad \sigma_x^2 = \mu \quad \sigma_x = \sqrt{\mu}$$

Here we see that both the mean and the variance of a Poisson random variable equal the average number of occurrences μ of the event of interest over the specified interval of time or space. For example, in the air traffic control situation, the Poisson distribution of x , the number of errors at the air traffic control center in a week, has a mean of $\mu_x = 4$ and a standard deviation of $\sigma_x = \sqrt{4} = .6325$.

FIGURE 4.9 Several Poisson Distributions



Exercises for Section 4.4



CONCEPTS

- 4.30 The values of a Poisson random variable are $x = 0, 1, 2, 3, \dots$. Explain what these values represent.
- 4.31 Explain the assumptions that must be satisfied when a Poisson distribution adequately describes a random variable x .

METHODS AND APPLICATIONS

- 4.32 Suppose that x has a Poisson distribution with $\mu = 2$.
- Write the Poisson formula and describe the possible values of x .
 - Starting with the smallest possible value of x , calculate $p(x)$ for each value of x until $p(x)$ becomes smaller than .001.
 - Graph the Poisson distribution using your results of b .
 - Find $P(x = 2)$.
 - Find $P(x \leq 4)$.
 - Find $P(x \geq 1)$ and $P(x > 2)$.
 - Find $P(1 \leq x \leq 4)$.
 - Find $P(2 < x < 5)$.
 - Find $P(2 \leq x < 6)$.
- 4.33 Suppose that x has a Poisson distribution with $\mu = 2$.
- Use the formulas given in this section to compute the mean, μ_x , variance, σ_x^2 , and standard deviation, σ_x .
 - Calculate the intervals $[\mu_x \pm 2\sigma_x]$ and $[\mu_x \pm 3\sigma_x]$. Then use the probabilities you calculated in Exercise 4.32 to find the probability that x will be inside each of these intervals.
- 4.34 A bank manager wishes to provide prompt service for customers at the bank's drive-up window. The bank currently can serve up to 10 customers per 15-minute period without significant delay. The average arrival rate is 7 customers per 15-minute period. Let x denote the number of customers arriving per 15-minute period. Assuming x has a Poisson distribution:
- Find the probability that 10 customers will arrive in a particular 15-minute period.
 - Find the probability that 10 or fewer customers will arrive in a particular 15-minute period.
 - Find the probability that there will be a significant delay at the drive-up window. That is, find the probability that more than 10 customers will arrive during a particular 15-minute period.
- 4.35 A telephone company's goal is to have no more than five monthly line failures on any 100 miles of line. The company currently experiences an average of two monthly line failures per 50 miles of line. Let x denote the number of monthly line failures per 100 miles of line. Assuming x has a Poisson distribution:
- Find the probability that the company will meet its goal on a particular 100 miles of line.
 - Find the probability that the company will not meet its goal on a particular 100 miles of line.
 - Find the probability that the company will have no more than five monthly failures on a particular 200 miles of line.
 - Find the probability that the company will have more than 12 monthly failures on a particular 150 miles of line.
- 4.36 A local law enforcement agency claims that the number of times that a patrol car passes through a particular neighborhood follows a Poisson process with a mean of three times per night shift. Let x denote the number of times that a patrol car passes through the neighborhood during a night shift.
- Calculate the probability that no patrol cars pass through the neighborhood during a night shift.
 - Suppose that during a randomly selected night shift no patrol cars pass through the neighborhood. Based on your answer in part a, do you believe the agency's claim? Explain.
 - Assuming that night shifts are independent and assuming that the agency's claim is correct, find the probability that exactly one patrol car will pass through the neighborhood on each of four consecutive nights.
- 4.37 When the number of trials, n , is large, binomial probability tables may not be available. Furthermore, if a computer is not available, hand calculations will be tedious. As an alternative, the Poisson distribution can be used to approximate the binomial distribution when n is large and p is small. Here the mean of the Poisson distribution is taken to be $\mu = np$. That is, when n is large and p is small, we can use the Poisson formula with $\mu = np$ to calculate binomial probabilities; we will obtain results close to those we would obtain by using the binomial formula. A common rule is to use this approximation when $np \geq 500$.
- To illustrate this approximation, in the movie *Conan*, a young female intern at a Boston hospital was very upset when her friend, a young nurse, went into a coma during routine anesthesia at the hospital. Upon investigation, she found that 10 of the last 30,000 healthy patients at the hospital had gone into comas during routine anesthesia. When she confronted the hospital administrator with this fact and the fact that the national average was 6 out of 100,000 healthy patients going

into comas during routine anesthesia, the administrator replied that 10 out of 30,000 was still quite small and thus not that unusual.

- Use the Poisson distribution to approximate the probability that 10 or more of 30,000 healthy patients would slip into comas during routine anesthesia, if in fact the true average at the hospital was 6 in 100,000. Hint: $\mu = np = 30,000(6/100,000) = 1.8$.

- Given the hospital's record and part a, what conclusion would you draw about the hospital's medical practices regarding anesthesia?

(Note: It turned out that the hospital administrator was part of a conspiracy to sell body parts and was purposely putting healthy adults into comas during routine anesthesia. If the intern had taken a statistics course, she could have avoided a great deal of danger.)

- 4.38 Suppose that an automobile parts wholesaler claims that .5 percent of the car batteries in a shipment are defective. A random sample of 200 batteries is taken, and four are found to be defective.
- Use the Poisson approximation discussed in Exercise 4.37 to find the probability that four or more car batteries in a random sample of 200 such batteries would be found to be defective, if we assume that the wholesaler's claim is true.
 - Based on your answer to part a, do you believe the claim? Explain.

Chapter Summary

In this chapter we began our study of random variables. We learned that a random variable represents an uncertain numerical outcome. We also learned that a random variable whose values can be listed is called a discrete random variable, while the values of a continuous random variable correspond to one or more intervals on the real number line. We saw that a probability distribution of a discrete random variable is a table, graph, or formula that gives the probability associated with each of the

random variable's possible values. We also discussed several descriptive measures of a discrete random variable—its mean (or expected value), its variance, and its standard deviation. We concluded this chapter by studying two important, commonly used discrete probability distributions—the binomial distribution and the Poisson distribution—and we demonstrated how these distributions can be used to make statistical inferences.

Glossary of Terms

binomial distribution: The probability distribution that describes a binomial random variable. (page 173)

binomial experiment: An experiment that consists of n independent, identical trials, each of which results in either a success or a failure and is such that the probability of success on any trial is the same. (page 173)

binomial random variable: A random variable that is defined to be the total number of successes in n trials of a binomial experiment. (page 173)

binomial tables: Tables in which we can look up binomial probabilities. (page 176)

continuous random variable: A random variable whose values correspond to one or more intervals of numbers on the real number line. (page 159)

discrete random variable: A random variable whose values can be counted or listed. (page 159)

expected value (of a random variable): The mean of the population of all possible observed values of a random variable. That is, the long-run average value obtained if values of a random variable are observed a (theoretically) infinite number of times. (page 163)

mean, variance, and standard deviation of a binomial random variable: page 180

Poisson probability formula: page 182

Mean, variance, and standard deviation of a Poisson random variable: page 185

Important Formulas

Properties of a discrete probability distribution: page 162

The mean (expected value) of a discrete random variable: page 163

Variance and standard deviation of a discrete random variable: page 166

Binomial probability formula: page 173

Supplementary Exercises



4.39 An investor holds two stocks, each of which can rise (R), remain unchanged (U), or decline (D) on any particular day. Let x equal the number of stocks that rise on a particular day.

a Write the probability distribution of x assuming that all outcomes are equally likely.

b Write the probability distribution of x assuming that for each stock $P(R) = .6$, $P(U) = .1$, and $P(D) = .3$ and assuming that movements of the two stocks are independent.

c Write the probability distribution of x assuming that for the first stock

$$P(R) = .4, P(U) = .2, P(D) = .4$$

and that for the second stock

$$P(R) = .8, P(U) = .1, P(D) = .1$$

and assuming that movements of the two stocks are independent.

4.40 Repeat Exercise 4.39, letting x equal the number of stocks that decline on the particular day.

4.41 Consider Exercise 4.39, and let x equal the number of stocks that rise on the particular day. Find μ_x and σ_x for

a The probability distribution of x in Exercise 4.39a.

b The probability distribution of x in Exercise 4.39b.

c The probability distribution of x in Exercise 4.39c.

d In which case is μ_x the largest? Interpret what this means in words.

e In which case is σ_x the largest? Interpret what this means in words.

4.42 Suppose that the probability distribution of a random variable x can be described by the formula

$$p(x) = \frac{(x-3)^2}{55}$$

for each of the values $x = -2, -1, 0, 1$, and 2 .

a Write the probability distribution of x .

b Show that the probability distribution of x satisfies the properties of a discrete probability distribution.

c Calculate the mean of x .

d Calculate the variance and standard deviation of x .

4.43 A rock concert promoter has scheduled an outdoor concert on July 4th. If it does not rain, the promoter will make \$30,000. If it does rain, the promoter will lose \$15,000 in guarantees made to the band and other expenses. The probability of rain on the 4th is .4.

a What is the promoter's expected profit? Is the expected profit a reasonable decision criterion? Explain.

b How much should an insurance company charge to insure the promoter's full losses? Explain your answer.

4.44 The demand (in number of copies per day) for a city newspaper is listed below with corresponding probabilities:

x = Demand	$p(x)$
90,000	.1
70,000	.25
90,000	.4
110,000	.2
130,000	.05

a Graph the probability distribution of x .

b Find the expected demand. Interpret this value, and label it on the graph of part a.

c Using Chebyshev's Theorem, find the minimum percentage of all possible daily demand values that will fall in the interval $[\mu_x \pm 2\sigma_x]$.

d Calculate the interval $[\mu_x \pm 2\sigma_x]$. Illustrate this interval on the graph of part a. According to the probability distribution of demand x previously given, what percentage of all possible daily demand values fall in the interval $[\mu_x \pm 2\sigma_x]$?

4.45 United Medicine, Inc., claims that a drug, Viro, significantly relieves the symptoms of a certain viral infection for 80 percent of all patients. Suppose that this drug is given to eight randomly selected patients who have been diagnosed with the viral infection.

a Let x equal the number of the eight randomly selected patients whose symptoms are significantly relieved. What distribution describes the random variable x ? Explain.

b Assuming that the company's claim is correct, find $P(x \leq 3)$.

c Suppose that of the eight randomly selected patients, three have had their symptoms significantly relieved by Viro. Based on the probability in part b, would you believe the claim of United Medicine, Inc.? Explain.

4.46 A consumer advocate claims that 80 percent of cable television subscribers are not satisfied with their cable service. In an attempt to justify this claim, a randomly selected sample of cable subscribers will be polled on this issue.

a Suppose that the advocate's claim is true, and suppose that a random sample of five cable subscribers is selected. Assuming independence, use an appropriate formula to compute the probability that four or more subscribers in the sample are not satisfied with their service.

b Suppose that the advocate's claim is true, and suppose that a random sample of 25 cable subscribers is selected. Assuming independence, find

(1) The probability that 15 or fewer subscribers in the sample are not satisfied with their service.

(2) The probability that more than 20 subscribers in the sample are not satisfied with their service.

(3) The probability that 20 and 24 (inclusive) subscribers in the sample are not satisfied with their service.

(4) The probability that exactly 24 subscribers in the sample are not satisfied with their service.

c Suppose that when we survey 25 randomly selected cable television subscribers, we find that 15 are actually not satisfied with their service. Using a probability you found in this exercise as the basis for your answer, do you believe the consumer advocate's claim? Explain.

4.47 A retail store has implemented procedures aimed at reducing the number of bad checks cashed by its cashiers. The store's goal is to cash no more than eight bad checks per week. The average number of bad checks cashed is three per week. Let x denote the number of bad checks cashed per week. Assuming that x has a Poisson distribution:

a Find the probability that the store's cashiers will not cash any bad checks in a particular week.

b Find the probability that the store will meet its goal during a particular week.

c Find the probability that the store will not meet its goal during a particular week.

d Find the probability that the store's cashiers will cash no more than 10 bad checks per two-week period.

e Find the probability that the store's cashiers will cash no more than five bad checks per three-week period.

4.48 Suppose that the number of accidents occurring in an industrial plant is described by a Poisson process with an average of 1.5 accidents every three months. During the last three months, four accidents occurred.

a Find the probability that no accidents will occur during the current three-month period.

b Find the probability that fewer accidents will occur during the current three-month period than occurred during the last three-month period.

c Find the probability that no more than 12 accidents will occur during a particular year.

d Find the probability that no accidents will occur during a particular year.

4.49 A high-security government installation has installed four security systems to detect attempted break-ins. The four security systems operate independently of each other, and each has a .85 probability of detecting an attempted break-in. Assume an attempted break-in occurs. Use the binomial distribution to find the probability that at least one of the four security systems will detect it.

4.50 A new stain removal product claims to completely remove the stains on 90 percent of all stained garments. Assume that the product will be tested on 20 randomly selected stained garments, and let x denote the number of these garments from which the stains will be completely removed. Use the binomial distribution to find $P(x \leq 13)$ if the stain removal product's claim is correct. If x actually turns out to be 13, what do you think of the claim?

4.51 Consider Exercise 4.50, and find $P(x \leq 17)$ if the stain removal product's claim is correct. If x actually turns out to be 17, what do you think of the claim?

4.52 A state has averaged one small business failure per week over the past several years. Let x denote the number of small business failures in the next eight weeks. Use the Poisson distribution to find $P(x \geq 17)$ if the mean number of small business failures remains what it has been. If x actually turns out to be 17, what does this imply?

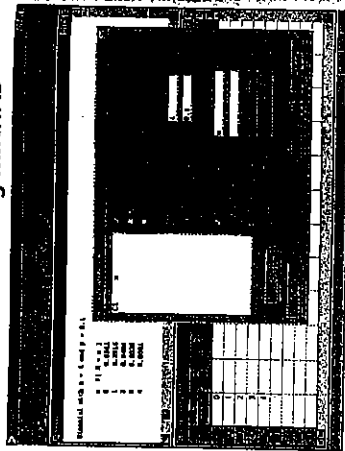
4.53 A candy company claims that its new chocolate almond bar averages 10 almonds per bar. Let x denote the number of almonds in the next bar that you buy. Use the Poisson distribution to find $P(x \leq 4)$ if the candy company's claim is correct. If x actually turns out to be 4, what do you think of the claim?

4.54 Consider Exercise 4.53, and find $P(x \leq 8)$ if the candy company's claim is true. If x actually turns out to be 8, what do you think of the claim?

Appendix 4.1 ■ Binomial and Poisson Probabilities Using MINITAB

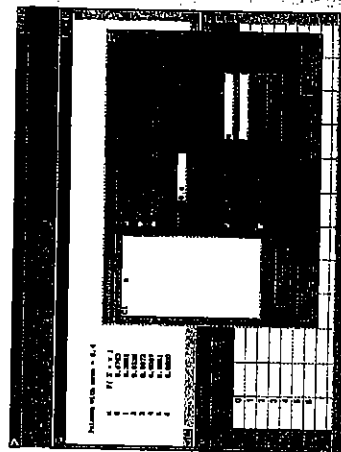
Binomial probabilities in Figure 4.4(a) on page 175;

- In the data window, enter the values 0 through 4 into column C1 and name the column x .
- Select Calc : Probability Distributions : Binomial.
- In the Binomial Distribution dialog box, select the Probability option by clicking.
- In the "Number of trials" box, enter 4 for the value of n .
- In the "Probability of success" box, enter 0.1 for the value of p .
- Select the "Input column" option and enter the variable name x into the Input column box.
- Click OK in the Binomial Distribution dialog box.
- The binomial probabilities will be displayed in the Session window.



Poisson probabilities in Figure 4.8(a) on page 184;

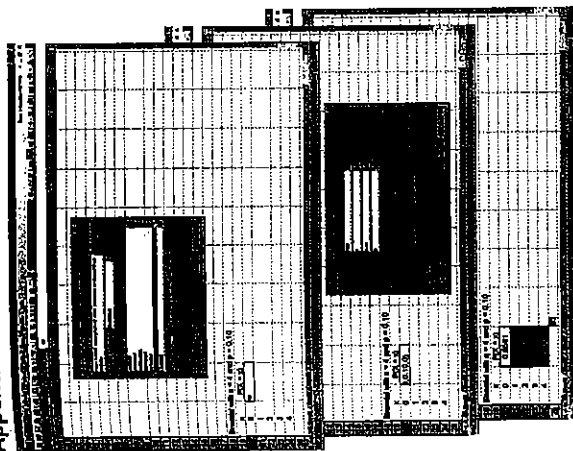
- In the data window, enter the values 0 through 6 into column C1 and name the column x .
- Select Calc : Probability Distributions : Poisson.
- In the Poisson Distribution dialog box, select the Probability option by clicking.
- In the Mean box, enter 0.4.
- Select the "Input column" option and enter the variable name x into the Input column box.
- Click OK in Poisson Distribution dialog box.
- The Poisson probabilities will be displayed in the Session window.



Appendix 4.2 ■ Binomial and Poisson Probabilities Using Excel

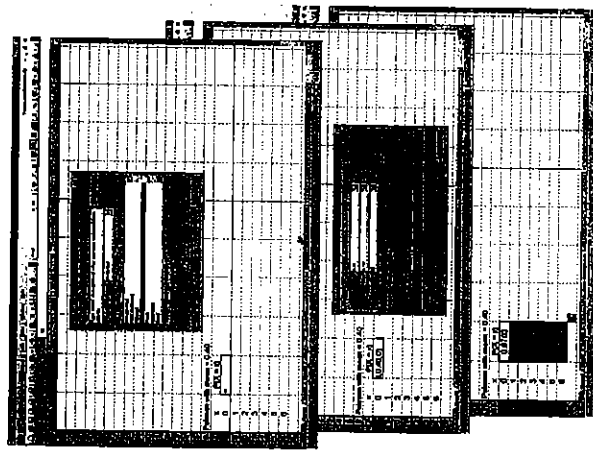
Binomial probabilities similar to Figure 4.4(a) on page 175:

- Enter the title, "Binomial with $n = 4$ and $p = 0.10$ ", in the location in which you wish to place the binomial results. Here we have placed the title beginning in cell A18 (any other choice will do).
- In cell A19, enter the heading, x .
- Enter the values 0 through 4 in cells A20 through A24.
- In cell B19, enter the heading $P(X = x)$.
- Click in cell B20 (this is where the first binomial probability will be placed). Click on the Insert Function button  on the Excel toolbar.
- In the Insert Function dialog box, select Statistical from the "Or select a category:" menu, select BINOMDIST from the "Select a function:" menu, and click OK.
- In the BINOMDIST Function Arguments dialog box, enter the cell location A20 (this cell contains the value for which the first binomial probability will be calculated) in the "Number_s" box.
- Enter the value 4 in the Trials box.
- Enter the value 0.10 in the "Probability_s" box.
- Enter the value 0 in the Cumulative box.
- Click OK in the BINOMDIST Function Arguments dialog box.
- When you click OK, the calculated result (0.6561) will appear in cell B20. Double-click the drag handle (in the lower right corner) of cell B20 to automatically extend the cell formula in B20 through cell B24.
- The remaining probabilities will be placed in cells B21 through B24.



Poisson probabilities similar to Figure 4.8(a) on page 194:

- Enter the title "Poisson with mean = 0.40" in the location in which you wish to place the Poisson results. Here we have placed the title beginning in cell A16 (any other choice will do).
- In cell A17, enter the heading, x .
- Enter the values 0 through 6 in cells A18 through A24.
- In cell B17, enter the heading, $P(X = x)$.
- Click in cell B18 (this is where the first Poisson probability will be placed). Click on the Insert Function button fx on the Excel toolbar.
- In the Insert Function dialog box, select Statistical from the "Or select a category" menu, select POISSON from the "Select a function" menu, and click OK.
- In the POISSON Function Arguments dialog box, enter the cell location A18 (this cell contains the value for which the first Poisson probability will be calculated) in the "X" box.
- Enter the value 0.40 in the Mean box.
- Enter the value 0 in the Cumulative box.
- Click OK in the POISSON Function Arguments dialog box.
- The calculated result for the probability of 0 events will appear in cell B18.
- Click on cell B18 and select Format : Cells.
- In the Format Cells dialog box, click on the Number tab, select Number from the Category menu, enter 4 in the Decimal places box, and click OK.
- The calculated result will now be rounded to 4 decimal places (0.6703). Double-click on the drag handle (in the lower right corner) of cell B18 to automatically extend the cell formula and formatting through cell B24.

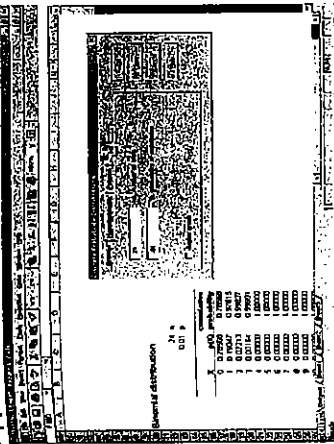


Appendix 4.3 ■ Binomial and Poisson Probabilities Using MegaStat

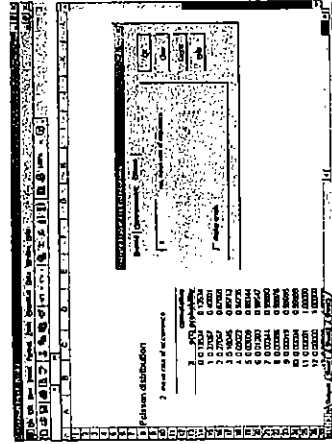
Binomial probabilities in Figure 4.7 on page 181.

- Select MegaStat : Probability : Discrete Probability Distributions.
- In the "Discrete Probability Distributions" dialog box, enter the number of trials (here equal to 24) and the probability of success p (here equal to .01) in the appropriate boxes. Click the Display Graph check box if a plot of the distribution is desired.
- Click OK in the "Discrete Probability Distributions" dialog box.

The binomial output is placed in an Output worksheet.



To calculate Poisson probabilities, click on the Poisson tab and enter the mean of the Poisson distribution. Then click OK.



To calculate Hypergeometric probabilities (discussed in part (2) of Appendix C), click on the Hypergeometric tab. Then enter the population size, the number of successes in the population, and the sample size in the appropriate boxes and click OK.